# p-adic Analysis 6

Siddharth K

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## Problem 3

Let the slopes of the Newton polygon of $f$ be $s_1, \ldots, s_m$ with corresponding lengths $l_1, \ldots, l_m$. By Theorem 6.2, the segment of slope $s_i$ and length $l_i$ means that $f(x)$ has $l_i$ roots with valuation $-s_i$. For each root $\alpha$, since $1/\alpha$ is also a root, $v_p(\alpha) = -v_p(1/\alpha)$ and each slope/length is paired with a negative slope and same length. So the segments are symmetric about the center.

## Problem 4

We have that

$$\binom{\sqrt{2}}{p} = \frac{\sqrt{2}(\sqrt{2}-1)\cdots(\sqrt{2}-p+1)}{p!}.$$

Since $v_p(1/p!) = -1$, $\binom{\sqrt{2}}{p} \in \mathbb{Z}_p[\sqrt{2}]$ if

$$v_p(\sqrt{2}(\sqrt{2}-1)\cdots(\sqrt{2}-p+1)) \geq 1.$$

Since each term in the product is equal to $\sqrt{2}-k$ for some $k \in \{0, 1, ..., p-1\}$, we need

$$\sqrt{2} - k \equiv 0 \pmod{p},$$

or

$$k^2 \equiv 2 \pmod{p},$$

for some $k$. So $\binom{\sqrt{2}}{p} \in \mathbb{Z}_p$ if such a $k$ exists.

## Problem 7

### Part (a)

Consider $x^2 - 2$. Since 2 is not a square modulo 5, the polynomial is irreducible, so the degree is 2 and the ramification index is 1.

## Part (b)

Consider $x^2 - 5$. This polynomial is Eisenstein at 5, so $x^2 - 5$ is irreducible is a totally ramified extension. So the degree is 2 and $e = 2$.